Game Projection and Robustness for Game-Theoretic Autonomous Driving

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***Abstract*— Game-theoretic decision making has the potential to bring human-like reasoning skills to autonomous vehicles (AVs), fostering trust between humans and AVs. However, to make these approaches sufficiently practical for real-world use, challenges such as *game complexity* and *incomplete information* have to be addressed. *Game complexity* refers to the difficulties in solving a game-theoretic problem, which include solution existence, algorithm convergence, and scalability. We show in our recent work that a possible solution to overcoming these difficulties is to use potential games. However, constructing a potential game often requires specific cost function designs, limiting their broad use. To address this challenge, we propose to employ a game projection technique in this paper, relaxing the cost function design conditions and making the potential game approach applicable to broader scenarios, even including the ones that cannot be modelled as a potential game. *Incomplete information* refers to the ego vehicle's lack of knowledge of other traffic agents' cost functions. In a driving scenario, deviations of the ego vehicle assumed/estimated others' cost functions from their actual ones are often inevitable. This necessitate the robustness analysis of a game-theoretic solution. This paper defines the robustness margin of a game solution as the maximum magnitude of cost function deviations that can be accommodated without changing the optimality of the game solution. With this definition, closed-form robustness margins are derived. Numerical studies using highway lane-changing scenarios are reported.**

***Index Terms*— Game theory, potential games, game decomposition, game projection, game robustness, autonomous driving.**

I. Introduction

FULLY autonomous vehicles (AVs) are envisioned to bring significant societal benefits, including improved road safety [2], increased fuel efficiency [3], reduced traffic congestion [4], enhanced mobility for the disabled and elderly [5], reduced need for parking space [6], smoother traffic flow [7], [8], and new business and research opportunities [9], [10]. However, major technical challenges still remain to be addressed before AVs can routinely drive on public roads, and one of them is the AV decision-making algorithm design [11], [12]. The AV decision-making algorithm serves as the AV's "brain" and directly determines the AV behavior, such as how the AV responds to the environment and/or to other road users including AVs, human-driven vehicles, bicycles, and pedestrians.

To enable interactive AVs with human-like reasoning skills and behavior, game-theoretic approaches have been explored in the literature [13], [14], [15], [16], [17], [18], [19], [20]. In game theory, agents are assumed to pursue self-interests while considering the interactions with others. Such a reasoning process is often considered consistent with humans' (and human drivers') reasoning and decision making [21], [22]. Along this line of research, both Nash games [17], [18] and Stackelberg games (also called leader-follower games) [19], [20] have been explored in the context of autonomous driving. However, these game-theoretic approaches often suffer from at least one of the two major challenges: *Game complexity* and *Incomplete information*.

Game complexity refers to the difficulties of solving a multi-player game. These difficulties include, but are not limited to, (a) Solution existence, i.e., the desired solution, e.g., pure strategy Nash equilibrium (PSNE), may not always exist; (b) Algorithm convergence, i.e., the solution-seeking algorithm, e.g., best or better response dynamics algorithms, may not always converge; and (c) Scalability, i.e., the computational complexity of solving a multi-player game can increase exponentially with the number of players [20], [23]. A few approaches/techniques have been developed to address some of these challenges. For example, to address the scalability challenge, pairwise games have been considered [17], [18], [19], [20]. However, such a pairwise setting may lead to conservative AV behavior and even deadlocks [24], as it only captures local interactions between a pair of agents as opposed to global interactions among all agents. To overcome this limitation, as well as to address the aforementioned challenges (a)-(c), a receding horizon potential game-based framework was developed in our previous work [1]. To construct a potential game, the cost function must be designed so that it satisfies certain conditions [1]. However, although the proposed cost function design can accommodate many driving objectives, there are still traffic scenarios/situations where

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agents' behavior cannot be modeled by the proposed cost function [17], [20], [25]. Therefore, approaches to extending the applicability of the potential game approach to situations where the cost function may not satisfy the conditions in [1], or even the game is not potential, are of significant interest.

Incomplete information, which is another major challenge in game-theoretic autonomous driving, refers to the ego vehicle's lack of global knowledge of other traffic agents' objective functions or cost functions. Agents' cost functions should reflect their driving preferences/habits/styles, which are often encoded as driver-specific parameters in the cost functions [19], [26]. To determine the values of these parameters, both offline calibration methods [26], [27], [28], [29], [30], [31] and online learning and estimation methods have been developed [19], [32]. However, irrespective of the approach, when facing specific agents on the road, there could be a mismatch between the estimated/learned and actual cost functions utilized by other agents [22]. Such a cost function deviation may also be caused by the limited sensing capabilities of the ego vehicle [33], [34], [35]. Due to this inevitable cost function deviation, it is important that the obtained game-theoretic solution has a certain level of robustness, such that the solution of the assumed game (i.e., the game with the ego vehicle estimated/learned cost functions of others) can still perform well and possibly even remain optimal, in the actual game (i.e., the game of agents' actual cost functions). Therefore, robustness analysis of game-theoretic autonomous driving policies needs to be conducted to find the robustness margins of a game-theoretic solution.

Motivated by the above considerations, this paper focuses on the development of a more ***practical*** and ***robust*** game-theoretic framework for autonomous driving. The contributions of this paper encompass two main aspects.

1) Towards practicality, this paper broadens the applicability of the potential game framework [1] to the case of non-potential games. This is achieved by exploiting the game decomposition techniques [36], [37] and by projecting an arbitrary game onto the potential game space. A game projection algorithm is provided.
2) Towards robustness, this paper defines the robustness margin of a game solution in autonomous driving and derives its closed-form expression. The robustness margin is defined as the maximum magnitude of cost function deviations such that the obtained PSNE retains its optimality in the original/actual game.

This paper is organized as follows. Section II introduces definitions and preliminaries. Section III formulates the AV decision-making problem as a game-theoretic problem. Section IV introduces the game decomposition approach to project an arbitrary game onto the potential game space. Section V establishes robustness analysis. Section VI reports the results from simulated autonomous driving case studies, and Section VII concludes the paper.

II. Definitions and Preliminaries

Let $\mathbb{Z}_+(\mathbb{Z}_{++})$ denote the set of non-negative (positive) integers and $\mathbb{R}_+(\mathbb{R}_{++})$ denote the set of non-negative (positive) real numbers.

To introduce game-theoretic background, we first define *strategic-form finite games*.

Definition 1 (Strategic-Form Finite Game [38]): A strategic-form finite game is defined as a tuple $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\}$, where

- $\mathcal{N} = \{1, 2, \cdots, N\}$ is a finite set of players or agents.
- $\mathcal{A} = \{\mathcal{A}_1, \mathcal{A}_2, \cdots, \mathcal{A}_N\}$ is a set of strategies with $\mathcal{A}_i$ representing the finite strategy space of player i.
- $J_i : \mathcal{A} \to \mathbb{R}$ is the cost function of player i.

Denote by $\mathcal{N}_{-i}$ the set of all agents except for agent i, i.e., $\mathcal{N}_{-i} = \{1, 2, \cdots, i-1, i+1, \cdots, N\}$. We let $\mathbf{a}_i \in \mathcal{A}_i$ represent the strategy of agent i and $\mathbf{a}_{-i} \in \mathcal{A}_{-i}$ represent the set of strategies of all other agents except for agent i, i.e., $\mathbf{a}_{-i} = \{\mathbf{a}_1, \ldots, \mathbf{a}_{i-1}, \mathbf{a}_{i+1}, \ldots, \mathbf{a}_N\}$ with $\mathcal{A}_{-i}$ being the domain of $\mathbf{a}_{-i}$. Denote $\mathbf{a} = \{\mathbf{a}_i, \mathbf{a}_{-i}\} \in \mathcal{A}$, which shall be referred to as a *strategy profile*. Denote by $h_i = |\mathcal{A}_i|$ the cardinality of the strategy space of player i, and $|\mathcal{A}| = \prod_{i=1}^N h_i$. Let $J_{-i} = \{J_1, \ldots, J_{i-1}, J_{i+1}, \ldots, J_N\}$ and $J = \{J_i, J_{-i}\}$.

We denote by $C_0 = \{f | f : \mathcal{A} \to \mathbb{R}\}$ the set of real-valued functions. The inner product in C_0 is defined [36] as:

$$\langle f_1, f_2 \rangle = \sum_{\mathbf{a}\in\mathcal{A}} f_1(\mathbf{a}) f_2(\mathbf{a}). \tag{1}$$

Given $\mathcal{N}$ and $\mathcal{A}$, a game is uniquely defined by its set of cost functions $\{J_i\}_{i\in\mathcal{N}} \in C_0^N$. Hence, the *space of games* with players $\mathcal{N}$ and strategy profiles $\mathcal{A}$, which is denoted by $\mathscr{G}_{\mathcal{N},\mathcal{A}}$, can be identified as $\mathscr{G}_{\mathcal{N},\mathcal{A}} \cong C_0^N$. Define the inner product on the space of games $\mathscr{G}_{\mathcal{N},\mathcal{A}}$ [36] as

$$\langle \mathcal{G}, \hat{\mathcal{G}} \rangle_{\mathcal{N},\mathcal{A}} \triangleq \sum_{i\in\mathcal{N}} h_i \langle J_i, \hat{J}_i \rangle, \tag{2}$$

where $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\}$ and $\hat{\mathcal{G}} = \{\mathcal{N}, \mathcal{A}, \{\hat{J}_i\}_{i\in\mathcal{N}}\}$.

The inner product (2) induces a norm to quantify the distance between games. Specifically, the norm on $\mathscr{G}_{\mathcal{N},\mathcal{A}}$ [36] is defined as

$$\|\mathcal{G}\|^2_{\mathcal{N},\mathcal{A}} = \langle \mathcal{G}, \mathcal{G} \rangle_{\mathcal{N},\mathcal{A}}. \tag{3}$$

Solution concepts in a non-cooperative game, including *Nash equilibrium* and *ϵ-Nash equilibrium*, are defined in Definitions 2-3.

Definition 2 (Pure-Strategy Nash Equilibrium [38]): An N-tuple of strategies (or strategy profile) $\{\mathbf{a}_1^*, \mathbf{a}_2^*, \ldots, \mathbf{a}_N^*\}$ is a pure-strategy Nash equilibrium for an N-player game if and only if

$$J_i(\mathbf{a}_i^*, \mathbf{a}_{-i}^*) \le J_i(\mathbf{a}_i, \mathbf{a}_{-i}^*) \quad \forall \mathbf{a}_i \in \mathcal{A}_i, \forall i \in \mathcal{N}. \tag{4}$$

Equation (4) implies that if a PSNE is achieved, then no player would have the incentive to change its strategy, representing a likely outcome for an interacting set of self-interested players such as drivers. Throughout this paper, we only consider pure strategies, and the term "pure strategy" may be omitted.

Definition 3 (ϵ-Nash Equilibrium [36]): Given $\epsilon \in \mathbb{R}_+$, a strategy profile $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_N\}$ is a pure-strategy ϵ-NE for an N-player game if and only if

$$J_i(\mathbf{a}_i, \mathbf{a}_{-i}) \le J_i(\mathbf{a}_i', \mathbf{a}_{-i}) + \epsilon \quad \forall \mathbf{a}_i' \in \mathcal{A}_i, \forall i \in \mathcal{N}. \tag{5}$$

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As shown by (4) and (5), both NE and ϵ-NE are defined in terms of cost differences at *comparable strategy profiles*, which are defined as a pair of strategy profiles that only differ in a single agent's strategy [36]. We denote the set of pairs of comparable strategy profiles by $\mathcal{C} \subset \mathcal{A} \times \mathcal{A}$, i.e., the strategy profiles $\mathbf{a} \in \mathcal{A}$ and $\mathbf{b} \in \mathcal{A}$ are comparable if $(\mathbf{a}, \mathbf{b}) \in \mathcal{C}$. A pair of comparable strategy profiles that differ in agent i's strategy is referred to as *i-comparable strategy profiles*. We denote the set of pairs of i-comparable strategy profiles by $\mathcal{C}_i \subset \mathcal{A} \times \mathcal{A}$, i.e., the strategy profiles $\mathbf{a} \in \mathcal{A}$ and $\mathbf{b} \in \mathcal{A}$ are i-comparable if $(\mathbf{a}, \mathbf{b}) \in \mathcal{C}_i$. Define a *pairwise comparison function* $X : \mathcal{A} \times \mathcal{A} \to \mathbb{R}$ as

$$X(\mathbf{a}, \mathbf{b}) = \begin{cases} J_i(\mathbf{a}) - J_i(\mathbf{b}), & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathcal{C}_i \text{ for some } i \in \mathcal{N}, \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Note that $X(\mathbf{a}, \mathbf{b})$ satisfies

$$X(\mathbf{a}, \mathbf{b}) = \begin{cases} -X(\mathbf{b}, \mathbf{a}), & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathcal{C}, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Such a pairwise comparison function results in a *flow representation* of a game [36] that emphasizes the change of costs between comparable strategy profiles. With the flow representation, each game is associated with a *game graph*, where the set of nodes corresponds to the strategy profiles, and the edges correspond to the cost differences of the deviating players in every comparable strategy profile pair. (See Figure 4 for an example.)

We denote by $C_1 = \{Y | Y : \mathcal{A} \times \mathcal{A} \to \mathbb{R}\}$ the set of functions that satisfy (7). In addition, we define an indicator function $W : \mathcal{A} \times \mathcal{A} \to \mathbb{R}$ as

$$W(\mathbf{a}, \mathbf{b}) = \begin{cases} 1, & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathcal{C}, \\ 0, & \text{otherwise.} \end{cases} \tag{8}$$

With the above notation, define the *combinatorial gradient operator* $\delta_0 : C_0 \to C_1$ as

$$(\delta_0 f)(\mathbf{a}, \mathbf{b}) = W(\mathbf{a}, \mathbf{b})(f(\mathbf{a}) - f(\mathbf{b})), \tag{9}$$

where $\mathbf{a}, \mathbf{b} \in \mathcal{A}$ and $f \in C_0$ [36].

Now we introduce player-specific operators. Let $W_i : \mathcal{A} \times \mathcal{A} \to \mathbb{R}$ be the indicator function for i-comparable strategy profiles given by

$$W_i(\mathbf{a}, \mathbf{b}) = \begin{cases} 1, & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathcal{C}_i, \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

It relates to W in (8) by $W = \sum_{i \in \mathcal{N}} W_i$.

Define the difference operator $D_i : C_0 \to C_1$ such that

$$(D_i f)(\mathbf{a}, \mathbf{b}) = W_i(\mathbf{a}, \mathbf{b})(f(\mathbf{a}) - f(\mathbf{b})), \tag{11}$$

where $\mathbf{a}, \mathbf{b} \in \mathcal{A}$ and $f \in C_0$ [36]. This operator quantifies the change in f between a pair of i-comparable strategy profiles. Relevant to D_i, we define $D : C_0^N \to C_1$ such that $D = [D_1, D_2, .., D_N]$.

Next we consider three classes of games, called *finite potential games*, *finite harmonic games*, and *finite nonstrategic games*. Throughout this paper, we only consider finite games, i.e., the games with a finite number of players and strategies, and the term "finite" may be omitted.

Definition 4 (Finite Potential Game [39]): The game $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i \in \mathcal{N}}\}$ is a finite potential game if and only if $\mathcal{N}$ and $\mathcal{A}$ contain a finite number of elements and a potential function $F : \mathcal{A} \to \mathbb{R}$ exists such that, $\forall i \in \mathcal{N}$,

$$J_i(\mathbf{a}_i, \mathbf{a}_{-i}) - J_i(\mathbf{a}_i', \mathbf{a}_{-i}) = F(\mathbf{a}_i, \mathbf{a}_{-i}) - F(\mathbf{a}_i', \mathbf{a}_{-i}), \quad \forall \mathbf{a}_i, \mathbf{a}_i' \in \mathcal{A}_i, \text{ and } \forall \mathbf{a}_{-i} \in \mathcal{A}_{-i}. \tag{12}$$

With the difference operator D_i defined in (11), the potential game can be alternatively defined as the game where a function $F : \mathcal{A} \to \mathbb{R}$ exists such that

$$D_i J_i = D_i F \tag{13}$$

holds for all $i \in \mathcal{N}$ [40].

Remark 1: Definition (4) suggests that potential games are games that allow the existence of a global function F, called the potential, such that the variations of J_i are fully characterized by the variations of F for all i. If we consider the flow representation of a game, then a potential game is a game where the total flow around every cycle in the game graph is always zero [36]. If a game is a non-potential game, then it must have a harmonic component that satisfies the flow conservation condition as illustrated in Definition 5.

Definition 5 (Finite Harmonic Game [36]): The game $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i \in \mathcal{N}}\}$ is a finite harmonic game if and only if $\mathcal{N}$ and $\mathcal{A}$ contain a finite number of elements and

$$\sum_{i \in \mathcal{N}} \sum_{\mathbf{a}_i' \in \mathcal{A}_i} \left(J_i(\mathbf{a}_i, \mathbf{a}_{-i}) - J_i(\mathbf{a}_i', \mathbf{a}_{-i}) \right) = 0, \quad \forall \mathbf{a}_i, \mathbf{a}_i' \in \mathcal{A}_i, \text{ and } \forall \mathbf{a}_{-i} \in \mathcal{A}_{-i}. \tag{14}$$

Eq. (14) indicates that harmonic games are games that satisfy flow conservation condition: For any strategy profile $(\mathbf{a}_i, \mathbf{a}_{-i})$, its total "flow", i.e., the sum of all pairwise comparisons $\sum_{i \in \mathcal{N}} \sum_{\mathbf{a}_i' \in \mathcal{A}_i} \left(J_i(\mathbf{a}_i, \mathbf{a}_{-i}) - J_i(\mathbf{a}_i', \mathbf{a}_{-i}) \right)$, equals zero. In contrast to potential games that always have a PSNE, a harmonic game does not have a PSNE, unless it is a nonstrategic game as defined in Definition 6.

Definition 6 (Finite Nonstrategic Game [41]): The game $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i \in \mathcal{N}}\}$ is a finite nonstrategic game if and only if $\mathcal{N}$ and $\mathcal{A}$ contain a finite number of elements and $\forall i \in \mathcal{N}$,

$$J_i(\mathbf{a}_i, \mathbf{a}_{-i}) = J_i(\mathbf{a}_i', \mathbf{a}_{-i}), \quad \forall \mathbf{a}_i, \mathbf{a}_i' \in \mathcal{A}_i, \text{ and } \forall \mathbf{a}_{-i} \in \mathcal{A}_{-i}. \tag{15}$$

Eq. (15) shows that for a nonstrategic game, every strategy profile is a PSNE. In addition, given two games $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i \in \mathcal{N}}\}$ and $\hat{\mathcal{G}} = \{\mathcal{N}, \mathcal{A}, \{\hat{J}_i\}_{i \in \mathcal{N}}\}$, they are strategically equivalent if the game $\{\mathcal{N}, \mathcal{A}, \{J_i - \hat{J}_i\}_{i \in \mathcal{N}}\}$ is a nonstrategic game [37]. Strategically equivalent games yield the same set of PSNE.

To define a representative for strategically equivalent games [36], we introduce a *normalized game* as follows.

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Definition 7 (Finite Normalized Game [36]): A game $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\}$ is called normalized if

$$\sum_{\mathbf{a}_i\in\mathcal{A}_i} J_i(\mathbf{a}_i, \mathbf{a}_{-i}) = 0, \quad \forall \mathbf{a}_{-i} \in \mathcal{A}_{-i}, \text{ and } \forall i \in \mathcal{N}. \tag{16}$$

As shown in [36], for any given game, there exists a unique strategically equivalent normalized game, i.e., these two games have identical pairwise comparison functions.

III. Problem Formulation

Consider a traffic scenario where multiple traffic agents $\mathcal{N} = \{1, 2, \cdots, N\}$ share the road (e.g., the lane-changing scenario depicted in Figure 1). Each agent aims to optimize its own driving performance, which is jointly affected by other road users so that the optimal strategies are determined according to

$$\begin{aligned}\mathbf{a}_i^*(t) &\in \operatorname*{argmin}_{\mathbf{a}_i(t)\in\mathcal{A}_i} J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}(t)) \\ &= \operatorname*{argmin}_{\mathbf{a}_i(t)\in\mathcal{A}_i} \sum_{\tau=t}^{t+T-1} \Psi_i(x_i(\tau), x_{-i}(\tau), a_i(\tau), a_{-i}(\tau)),\end{aligned} \tag{17}$$

where $t \in \mathbb{Z}_+$ is the time index, $\mathbf{a}_i(t) = \{a_i(t), a_i(t+1), \ldots, a_i(t+T-1)\} \in \mathcal{A}_i$ is agent i's strategy, i.e., agent i's action sequence over the horizon $[t, t+1, \ldots, t+T-1]$, $T \in \mathbb{Z}_{++}$ is the prediction horizon length, $a_i(t) \in \mathcal{U}_i$ is agent i's action at t, which can represent, for instance, a traffic agent's acceleration and/or steering angle, $x_i(t) \in \mathcal{X}_i$ is agent i's state, which typically includes the position and velocity information, $\Psi_i : \mathcal{A} \times \mathcal{X} \to \mathbb{R}$, where $\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_N\}$, is agent i's instantaneous cost at one time instant, and $J_i : \mathcal{A} \to \mathbb{R}$ is agent i's cumulative cost over the T steps horizon. Note that the expression of $J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}(t))$ can be different at different t due to its state dependence, i.e., dependence on states $x_i(t)$ and $x_{-i}(t)$. With a slight abuse of notation, we do not distinguish J_i at different t, as this state dependence is clear from the context. After deriving $\mathbf{a}_i^*(t)$, agent i implements the first element $a_i^*(t)$ and repeats the same procedure at $t+1$ with a shifted horizon.

We use a general nonlinear function g_i to represent agent i's state evolution model:

$$x_i(t+1) = g_i(x_i(t), a_i(t)). \tag{18}$$

Examples of g_i include a single-mass model [42], [43], a unicycle model [44], and a bicycle model [45]. For more detailed discussions on the physical interpretations of x_i, a_i, g_i, and J_i, and the rationality of the receding horizon game (17) in the autonomous driving context, please refer to [22].

Such a receding horizon game formulation facilitates predictive capability for the AV, allows fast response to changes in the environment, and tackles agents' interactions from a self-interest optimization perspective, consistent with human driving experience. However, solving such a multi-player game (17) is not straightforward, and the challenges include

1) ***Existence of solution:*** The strategy profile $\{\mathbf{a}_1^*(t), \mathbf{a}_2^*(t), \ldots, \mathbf{a}_N^*(t)\}$ that satisfies (17) for all $i \in \mathcal{N}$ may not always exist.

Algorithm 1 Best Response Dynamics to Solve (17)

Inputs:
Agent set $\mathcal{N}$;
System state $x(t)$;
System dynamics g_i, $i \in \mathcal{N}$;
Strategy space $\mathcal{A}$;
Cost functions J_i, $i \in \mathcal{N}$.
Output:
PSNE $\mathbf{a}^*(t)$.
Procedures:
Initialize $\mathbf{a}(t) \in \mathcal{A}$
Set *NashCondition=False*
While *NashCondition=False* **do**
  For $i = 1, 2, \ldots, N$ **do**
    Find $\mathbf{a}_i^*(t)$ according to
$$\mathbf{a}_i^*(t) \in \operatorname*{argmin}_{\mathbf{a}_i(t)\in\mathcal{A}_i} J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}(t)).$$
    Update $\mathbf{a}_i(t)$ using $\mathbf{a}_i^*(t)$.
  End for
  If
    $\mathbf{a}_i^*(t) \in \operatorname*{argmin}_{\mathbf{a}_i(t)\in\mathcal{A}_i} J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}^*(t))$ holds $\forall i \in \mathcal{N}$,
  Then
    Set *NashCondition=True*.
  End if
End while

2) ***Convergence of algorithm:*** A solution seeking algorithm, e.g., best- or better- response dynamics algorithm, may not always converge.
3) ***Nonuniqueness of PSNE:*** The game (17) may yield multiple solutions, i.e., multiple PSNE. Selecting the most preferable one is not straightforward.
4) ***Computational scalability:*** Solving (17) is often computationally expensive when N is large. Specifically, if using best response dynamics (i.e., Algorithm 1), then the number of required optimizations can increase exponentially with the number of agents [23].
5) ***Lack of information:*** To solve (17), the ego vehicle needs to know all agents' cost functions $\{J_i\}_{i\in\mathcal{N}}$. This information is not available in real-world traffic scenarios considering the variability of human driving styles.

To solve the challenges $1-4$, a receding horizon potential game framework was developed in our recent studies [1]. As detailed in [1], if the game (17) is a potential game, then the following properties hold:

- The game (17) has at least one PSNE.
- Algorithm 1 always converges.
- One can find a PSNE that is both individually optimal (in the sense of self-interest optimization, i.e., NE) and socially optimal (in the sense of social interest optimization with "social interest" characterized by the potential function). Such a PSNE is often preferable to other solutions as it represents a considerate AV which cares about not only itself but also other traffic agents.

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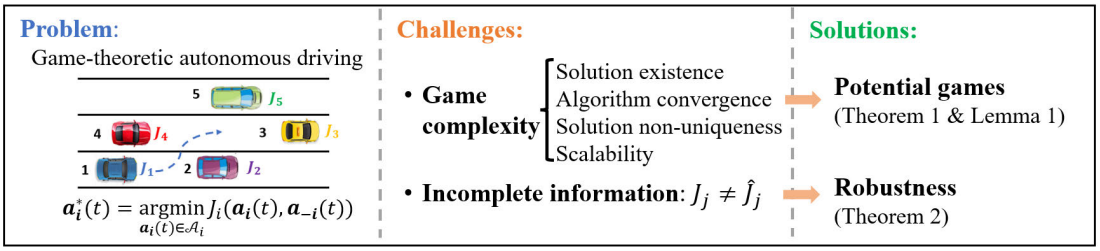

Fig. 1. Overview of the problem.

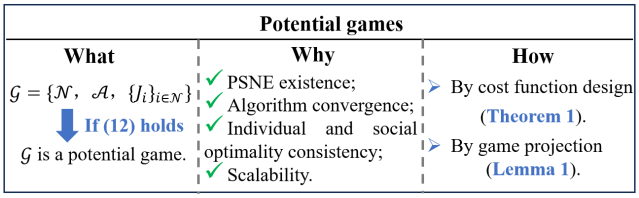

Fig. 2. Potential games: What, Why, and How.

- The computational challenge is addressed by the potential function optimization algorithm (Algorithm 2), since this algorithm only requires *one* optimization procedure regardless of the number of agents N.

Despite the aforementioned merits, it is often not straightforward to make the game (17) a potential game. It is shown in [1] that if agents' cost functions are designed so that they satisfy certain conditions, then the resulting game is a potential game. We include this result in Theorem 1.

Theorem 1 (Theorem 6 in [1]): If the cost function J_i in (17) satisfies

$$\begin{aligned} &J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}(t)) \\ &= \alpha J_i^{self}(\mathbf{a}_i(t)) + \beta \sum_{j\in\mathcal{N}, j\neq i} J_{ij}(\mathbf{a}_i(t), \mathbf{a}_j(t)), \end{aligned} \tag{19}$$

where $J_i^{self} : \mathcal{A}_i \to \mathbb{R}$ is a function determined solely by agent i's strategy, $J_{ij} : \mathcal{A}_i \times \mathcal{A}_j \to \mathbb{R}$ satisfies

$$\begin{aligned} &J_{ij}(\mathbf{a}_i(t), \mathbf{a}_j(t)) \\ &= J_{ji}(\mathbf{a}_j(t), \mathbf{a}_i(t)), \\ &\quad \forall i, j \in \mathcal{N}, i \neq j, \text{ and } \forall \mathbf{a}_i(t) \in \mathcal{A}_i, \forall \mathbf{a}_j(t) \in \mathcal{A}_j, \end{aligned} \tag{20}$$

and α and β are two real numbers. Then the game (17) is a potential game with the following potential function,

$$\begin{aligned} &F(\mathbf{a}(t)) \\ &= \alpha \sum_{i\in\mathcal{N}} J_i^{self}(\mathbf{a}_i(t)) + \beta \sum_{i\in\mathcal{N}} \sum_{j\in\mathcal{N}, j<i} J_{ij}(\mathbf{a}_i(t), \mathbf{a}_j(t)). \end{aligned} \tag{21}$$

Theorem 1 states that if J_i in (17) satisfies (19), then the game (17) is a potential game. For the cost (19), the first term J_i^{self} can be used to model self-centered objectives, including tracking a desired speed, optimizing ride comfort, maximizing fuel efficiency, and so on. The second term J_{ij} can be used to model symmetric pairwise interactions, such as a pairwise collision penalty [1]. For specific cost function design examples satisfying (19) and the interpretations in various traffic scenarios, please refer to our recent studies [1], [22].

Algorithm 2 Potential Function Optimization to Solve (17)

Input:
Agent set $\mathcal{N}$;
System state $x(t)$;
System dynamics g_i, $i \in \mathcal{N}$;
Strategy space $\mathcal{A}$;
Cost functions J_i, $i \in \mathcal{N}$, that satisfy (19).

Output:
PSNE $\mathbf{a}^*(t)$.

Procedures:
1: Find $F(\mathbf{a}(t))$ according to (21).
2: Find $\mathbf{a}^*(t)$ such that

$$\mathbf{a}^*(t) = \underset{\mathbf{a}(t)\in\mathcal{A}}{\operatorname{argmin}}\ F(\mathbf{a}(t)). \tag{22}$$

Although the cost function design in (19)-(20) can accommodate many driving objectives, there may still exist traffic scenarios where agents' interactions are asymmetric. This asymmetry may be caused by road priority rules or by significant vehicle size differences such as the interaction between a sedan and a large truck in highway merging. The modeling of these asymmetric interactions could result in J_{ij} not satisfying (20), invalidating the conditions of Theorem 1. We shall develop techniques in Section IV such that the potential game approach is applicable even if agents' cost functions do not satisfy (19).

In addition, as described in Challenge 5, due to the lack of information, the ego vehicle can only rely on the assumed cost functions of other traffic agents, leading to cost function deviations between the game solved by the ego vehicle and the game of agents' actual costs. Quantification of whether and how such cost function deviations affect the game solution is of interest and shall be addressed in Section V. (See Figure 1 for an overview of the problem and Figure 2 for the use of potential games.)

IV. Game Projection

To expand the use of the potential game framework to address games where agents' cost functions may not satisfy (19)-(20), or which could be non-potential to begin with, we propose to use game decomposition technique. This technique was developed based on Helmholtz decomposition and on the flow representation of a game [36]. The main result is included in Lemma 1.

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Lemma 1 (Theorem 4.1 in [36]): The space of games $\mathscr{G}_{\mathcal{N},\mathcal{A}}$ is a direct sum of the potential, harmonic, and nonstrategic subspaces, i.e.,

$$\mathscr{G}_{\mathcal{N},\mathcal{A}} = \mathscr{P} \oplus \mathscr{H} \oplus \mathscr{N}, \tag{23}$$

where the potential subspace $\mathscr{P}$ is the set of games that satisfy both (12) and (16), the harmonic subspace $\mathscr{H}$ is the set of games that satisfy both (14) and (16), and nonstrategic subspace $\mathscr{N}$ is the set of games that satisfy (15).

In particular, given a game $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\}$, it can be uniquely decomposed into three components:

- *Potential component: $J^P \triangleq D^\dagger \delta_0 \delta_0^\dagger D J$,*
- *Harmonic component: $J^H \triangleq D^\dagger (I - \delta_0 \delta_0^\dagger) D J$,*
- *Nonstrategic component: $J^N \triangleq (I - D^\dagger D) J$,*

where $J = \{J_i\}_{i\in\mathcal{N}}$, $J^P + J^H + J^N = J$, $\delta_0^\dagger$ is the (Moore-Penrose) pseudoinverse of the operator δ_0 (defined in (9)) with respect to the inner product in (1), and $D^\dagger$ is the pseudoinverse of the operator D (defined under (11)) with respect to the inner product in C_0^N, which is defined as the sum of the inner products in all C_0 components.

Lemma 1 shows that the space of games can be decomposed into three subspaces: $\mathscr{P}$, $\mathscr{H}$, and $\mathscr{N}$, and these three subspaces are orthogonal under the inner product defined in (2). Note that the direct sum of $\mathscr{P}$ and $\mathscr{N}$, i.e., $\mathscr{P} \oplus \mathscr{N}$, coincides with the set of all potential games, i.e., the games that satisfy (12), and the direct sum of $\mathscr{H}$ and $\mathscr{N}$, i.e., $\mathscr{H} \oplus \mathscr{N}$, coincides with the set of all harmonic games, i.e., the games that satisfy (14). That is:

$$\mathscr{G}_{\mathcal{N},\mathcal{A}} = \underbrace{\mathscr{P} \quad \oplus \quad \overbrace{\mathscr{N} \quad \oplus \quad \mathscr{H}}^{\text{Harmonic games}}}_{\text{Potential games}}$$

With Lemma 1, given an arbitrary game $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$, one can project it onto the subspace $\mathscr{P} \oplus \mathscr{N} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$ to find its "closest" potential game with respect to the inner product defined in (2) to best approximate it. We denote the projected game as $\hat{\mathcal{G}} = \{\mathcal{N}, \mathcal{A}, \{\hat{J}_i\}_{i\in\mathcal{N}}\} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$. Such a projection process can be realized by solving a least-square optimization problem:

$$\begin{aligned} &\min_{F, \hat{J}_i} \sum_{i\in\mathcal{N}} h_i \| J_i - \hat{J}_i \|^2 \\ &s.t. \quad D_i \hat{J}_i = D_i F, \quad \forall i \in \mathcal{N}. \end{aligned} \tag{24}$$

where F is the potential function of the potentialized game, D_i is the difference operator defined in (11), and the constraint in (24) is to enforce the potential game condition as specified in (13). The detailed algorithm for projecting an arbitrary game onto the potential game space is provided in Algorithm 3.

The above game projection approach provides a possible solution to approximating any game, which may or may not be a potential game, using its closest potential game. If the original game is a potential game, then the projected game is the same as the original. If the original game is not a potential game, then the resulting game differs from the original in terms of agents' costs, i.e., $\exists i \in \mathcal{N}$, such that $J_i \neq \hat{J}_i$. Such a cost function deviation may lead to the solution deviation, i.e., the PSNE in the potentialized game may not be a PSNE in the original game. The robustness analysis in Section III assesses the impact of cost function deviation on the solution deviation.

Algorithm 3 Potentialize a Game

Input:
The game to potentialize $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\}$.

Output:
Potentialized game: $\hat{\mathcal{G}} = \{\mathcal{N}, \mathcal{A}, \{\hat{J}_i\}_{i\in\mathcal{N}}\}$ with the potential function F.

Procedures:
1: Find $\hat{J}_i(\mathbf{a})$ and $F(\mathbf{a})$ $\forall \mathbf{a} \in \mathcal{A}$ and $\forall i \in \mathcal{N}$ by solving the following least-squares problem:

$$\begin{aligned} &\min_{F(\mathbf{a}), \hat{J}_i(\mathbf{a})} \sum_{i\in\mathcal{N}} h_i \| J_i(\mathbf{a}) - \hat{J}_i(\mathbf{a}) \|^2 \\ &s.t. \quad (D_i F)(\mathbf{a}, \mathbf{b}) = (D_i \hat{J}_i)(\mathbf{a}, \mathbf{b}), \quad \forall i \in \mathcal{N}, \forall \mathbf{a}, \mathbf{b} \in \mathcal{A}. \end{aligned} \tag{25}$$

V. Robustness Analysis

This section considers the robustness margin of a game if perturbations exist in agents' cost functions. These perturbations may be caused by the game projection as described in Section IV, or by the ego vehicle's incomplete information of other agents' cost functions as specified in Challenge 5 in Section III. We do not distinguish the sources of the perturbations here, but are only interested in the effect of the perturbations on the optimality of the solution. This part of the study is inspired by [46], but we use different robustness margin definitions and analysis to meet our specific aims.

Consider two games $\mathcal{G} = \{\mathcal{N}, \mathcal{A}, \{J_i\}_{i\in\mathcal{N}}\} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$ and $\hat{\mathcal{G}} = \{\mathcal{N}, \mathcal{A}, \{\hat{J}_i\}_{i\in\mathcal{N}}\} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$. Let $\hat{\mathcal{G}}$ represent the game solved by the ego vehicle. Then the game $\mathcal{G}$ can represent the original game before the game projection and/or the game associated with other agents' actual cost functions unknown to the ego vehicle. Define $\Delta J_i : \mathcal{A} \to \mathbb{R}$ as

$$\Delta J_i = J_i - \hat{J}_i. \tag{26}$$

Definition 8 (Robustness Margin): Consider the games $\mathcal{G}$ and $\hat{\mathcal{G}}$. Let ΔJ_i be given by (26). The robustness margin of a PSNE $\hat{\mathbf{a}}^*$ solved from the game $\hat{\mathcal{G}}$ (denoted as $\mu(\hat{\mathbf{a}}^*)$) is defined as the supremum of the magnitude $\|\Delta J\|_\infty$ such that $\hat{\mathbf{a}}^*$ is also a PSNE of the game $\mathcal{G}$, where

$$\|\Delta J\|_\infty = \max_{i\in\mathcal{N}} \|\Delta J_i\|_\infty, \tag{27}$$

$$\|\Delta J_i\|_\infty = \max_{\mathbf{a}\in\mathcal{A}} |\Delta J_i(\mathbf{a})|. \tag{28}$$

Theorem 2: Consider the games $\mathcal{G}$ and $\hat{\mathcal{G}}$ and the robustness margin defined in Definition 8. Then the robustness margin $\mu(\hat{\mathbf{a}}^*)$ is

$$\mu(\hat{\mathbf{a}}^*) = \frac{1}{2} \min_{i\in\mathcal{N}} \chi_i(\hat{\mathbf{a}}^*), \tag{29}$$

where $\chi_i : \mathcal{A} \to \mathbb{R}$ is given by

$$\chi_i(\mathbf{a}) = \min_{(\mathbf{a},\mathbf{b})\in\mathcal{C}_i, \mathbf{a}\neq\mathbf{b}} \left(\hat{J}_i(\mathbf{b}) - \hat{J}_i(\mathbf{a}) \right). \tag{30}$$

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Proof: To show that (29) holds, in the first step, we show that

$$\mu(\hat{\mathbf{a}}^*) \geq \frac{1}{2}\min_{i\in\mathcal{N}} \chi_i(\hat{\mathbf{a}}^*). \tag{31}$$

In the second step, we show that

$$\mu(\hat{\mathbf{a}}^*) \leq \frac{1}{2}\min_{i\in\mathcal{N}} \chi_i(\hat{\mathbf{a}}^*). \tag{32}$$

Step 1: To prove (31).

For every $\{\Delta J_i\}_{i\in\mathcal{N}}$ that satisfies $\|\Delta J\|_\infty \leq \frac{1}{2}\min_{i\in\mathcal{N}} \chi_i(\hat{\mathbf{a}}^*)$, it must satisfy

$$\|\Delta J_i\|_\infty \leq \frac{1}{2}\chi_i(\hat{\mathbf{a}}^*), \quad \forall i \in \mathcal{N}. \tag{33}$$

Eq. (33) leads to

$$\Delta J_i(\hat{\mathbf{a}}^*) - \Delta J_i(\mathbf{b}) \leq 2\|\Delta J_i\|_\infty \leq \hat{J}_i(\mathbf{b}) - \hat{J}_i(\hat{\mathbf{a}}^*), \quad \forall \mathbf{b} \neq \hat{\mathbf{a}}^* \text{ and } (\hat{\mathbf{a}}^*, \mathbf{b}) \in \mathcal{C}_i. \tag{34}$$

Therefore, the following inequality holds:

$$\Delta J_i(\hat{\mathbf{a}}^*) + \hat{J}_i(\hat{\mathbf{a}}^*) \leq \Delta J_i(\mathbf{b}) + \hat{J}_i(\mathbf{b}). \tag{35}$$

Combining (26) and (35), we have

$$J_i(\hat{\mathbf{a}}^*) \leq J_i(\mathbf{b}). \tag{36}$$

Because (36) holds $\forall i \in \mathcal{N}$ and $\forall \mathbf{b} \neq \hat{\mathbf{a}}^*$, $(\hat{\mathbf{a}}^*, \mathbf{b}) \in \mathcal{C}_i$, according to Definition 2, $\hat{\mathbf{a}}^*$ is a PSNE for the game $\mathcal{G}$, completing the proof of (31).

Step 2: To prove (32).

Let us construct a $\{\Delta J_i\}_{i\in\mathcal{N}}$ with $\|\Delta J\|_\infty = \frac{1}{2}\min_{i\in\mathcal{N}} \chi_i(\hat{\mathbf{a}}^*) + \epsilon$, where $\epsilon > 0$ is an arbitrary small number. We show that with the constructed $\{\Delta J_i\}_{i\in\mathcal{N}}$, the strategy profile $\hat{\mathbf{a}}^*$ is not a PSNE of the game $\mathcal{G}$, and thus, the inequality (32) holds.

Consider the game $\hat{\mathcal{G}}$. Let $j \in \mathcal{N}$ be the agent that achieves the maximum on the right-hand side of (27). Let $\mathbf{b} \in \mathcal{A}$ be the startegy profile that achieves the minimum on the right-hand side of (30) with $\mathbf{a}$ being $\hat{\mathbf{a}}^*$. Construct a $\{\Delta J_i\}_{i\in\mathcal{N}}$ such that

$$\Delta J_j(\hat{\mathbf{a}}^*) = \frac{1}{2}\left(\hat{J}_j(\mathbf{b}) - \hat{J}_j(\hat{\mathbf{a}}^*)\right) + \epsilon, \tag{37}$$

$$\Delta J_j(\mathbf{b}) = -\Delta J_j(\hat{\mathbf{a}}^*). \tag{38}$$

$$\Delta J_i(\mathbf{a}) = 0, \forall i \neq j, \text{ and } \forall \mathbf{a} \notin \{\hat{\mathbf{a}}^*, \mathbf{b}\}. \tag{39}$$

Equations (37) and (38) lead to

$$\Delta J_j(\hat{\mathbf{a}}^*) - \Delta J_j(\mathbf{b}) = 2\Delta J_j(\hat{\mathbf{a}}^*) = \left(\hat{J}_j(\mathbf{b}) - \hat{J}_j(\hat{\mathbf{a}}^*)\right) + 2\epsilon. \tag{40}$$

Therefore, we have

$$\Delta J_j(\hat{\mathbf{a}}^*) + \hat{J}_j(\hat{\mathbf{a}}^*) = \Delta J_j(\mathbf{b}) + \hat{J}_j(\mathbf{b}) + 2\epsilon. \tag{41}$$

Combining (26) and (41), we have

$$J_j(\hat{\mathbf{a}}^*) = J_j(\mathbf{b}) + 2\epsilon. \tag{42}$$

Thus,

$$J_j(\mathbf{b}) < J_j(\hat{\mathbf{a}}^*). \tag{43}$$

Note that $(\hat{\mathbf{a}}^*, \mathbf{b}) \in \mathcal{C}_j$. Therefore, $\hat{\mathbf{a}}^*$ is not a PSNE of the game $\mathcal{G}$, completing the proof of (32).

The inequalities (31) and (32) together lead to (29), completing the proof of Theorem 2. □

With the robustness margin in Theorem 2, after projecting a game onto the potential game space using Algorithm 3 and deriving the PSNE of the projected game using Algorithm 2, one may check whether $\|\Delta J\|_\infty \leq \frac{1}{2}\min_{i\in\mathcal{N}} \chi_i(\hat{\mathbf{a}}^*)$ holds. If yes, then this NE is also a NE of the original game, despite that the original game may not be a potential game. We shall provide a specific numerical example in the next section.

Remark 2: Theorem 2 provides a possible solution to quantifying how "trustworthy" a game-theoretic solution is in the presence of incomplete information, which refers to the ego vehicle lack of knowledge of other traffic agents' cost functions. This lack of knowledge can be due to unknown human driving styles, sensing/perception errors, and/or the ego vehicle limited sensing distances. These factors may all lead to the situation in which $\hat{J}_{-i} \neq J_{-i}$. However, as we show in Theorem 2, the cost function deviation may not necessarily lead to a solution deviation. That is, a solution to the "assumed game" with $\hat{J}$ may retain its optimality in the "actual game" with J. Whether a solution deviation happens depends on the robustness margin of the game. If a game solution has a large robustness margin, then a broad range of agents' costs can be accommodated without loosing the solution optimality, indicating a trustworthy solution.

Theorem 2 also reveals that the robustness margin of a PSNE depends on $\min_{(\hat{\mathbf{a}}^*,\mathbf{b})\in\mathcal{C}_i, \hat{\mathbf{a}}^*\neq\mathbf{b}} \left(\hat{J}_i(\mathbf{b}) - \hat{J}_i(\hat{\mathbf{a}}^*)\right)$. *Intuitively, it indicates that for given cost functions (e.g., (45)) and strategy range (e.g., feasible acceleration range), the sparser the strategy space is, the more robust the game solution is. For example, if a vehicle's strategy is its acceleration, then the game with the strategy space* $[-2, 0, 2]$ m/s^2 *would result in a larger robustness margin than that with* $[-3, -2, -1, 0, 1, 2, 3]$ m/s^2. *This result suggests that game-theoretic approaches might be more reliable (in the sense of a larger robustness margin) in anticipating other agents' high-level intents (e.g., whether a vehicle would yield or proceed) compared to predicting their accurate maneuvers (e.g., whether it brakes at* -2 m/s^2 *or* -1 m/s^2*). This observation is consistent with human driving experience. When a human driver interacts with others, e.g., to merge onto a highway, he/she often "predicts" whether a vehicle has the intent to yield, as opposed to predicting their precise accelerations.*

In the case where the maximum magnitude of the cost function deviation exceeds the robustness margin in (29), the PSNE of $\mathcal{G}$ may not necessarily be a PSNE of $\hat{\mathcal{G}}$ any more, but it can still be an ϵ-NE in $\hat{\mathcal{G}}$ (see Definition 5 for ϵ-NE). The value of ϵ depends on the distance of the two games and is given by the following lemma.

Lemma 2 (Theorem 6.3 in [36]): Let $\mathcal{G} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$ *and* $\hat{\mathcal{G}} \in \mathscr{G}_{\mathcal{N},\mathcal{A}}$ *be two games. Define* $\alpha \triangleq \|\mathcal{G} - \hat{\mathcal{G}}\|_{\mathcal{N},\mathcal{A}}$. *Then every PSNE of* $\hat{\mathcal{G}}$ *is an* ϵ*-NE of* $\mathcal{G}$ *for some* $\epsilon \leq \max_{i\in\mathcal{N}} \frac{2\alpha}{\sqrt{h_i}}$.

Lemma 2 indicates that although the cost function deviation may lead to a solution deviation, we can quantify the optimality of the obtained solution in the original game.

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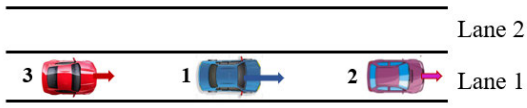

Fig. 3. Three-vehicle highway lane-changing scenario.

VI. Numerical Studies

This section reports the outcome from numerical studies to illustrate and verify the algorithms and analysis in Sections III-V. The simulation animation videos are available at *https://youtu.be/BHmX-ldwWOU*.

A. Scenario Setup

We consider a highway lane-changing scenario as pictured in Figure 3. The vehicles' dynamics are described by a kinematic bicycle model [45], [47]:

$$\begin{aligned} x_i(t+1) &= x_i(t) + v_i(t)\cos(\phi_i(t)+\beta_i(t))\Delta t, \\ y_i(t+1) &= y_i(t) + v_i(t)\sin(\phi_i(t)+\beta_i(t))\Delta t, \\ v_i(t+1) &= v_i(t) + w_i(t)\Delta t, \\ \phi_i(t+1) &= \phi_i(t) + \frac{v_i(t)}{l_r}\sin(\beta_i(t))\Delta t, \\ \beta_i(t+1) &= \beta_i(t) + \tan^{-1}\left(\frac{l_r}{l_r+l_f}\tan(\delta_{i,f}(t))\right)\Delta t, \end{aligned} \tag{44}$$

where $i = 1, 2, 3$ designates the ith vehicle; $\Delta t = 0.5s$ is the sampling time; x_i and y_i are the longitudinal and lateral position of the center of mass of vehicle i along x and y axes, respectively; v_i and w_i are the velocity and acceleration of the center of mass of vehicle i, respectively; β_i is the angle of the velocity with respect to the longitudinal axis of the vehicle; ϕ_i is the inertial heading; l_f and l_r are the lengths from the center of mass to the front and rear ends of the car, respectively, with $l_f = l_r = 1.5\ m$ and with vehicle width being $1.5\ m$; $\delta_{i,f}$ is the steering angle of the front wheels, and the rear wheel steering angles are assumed to be zero since for many vehicles the rear wheels cannot be steered [45]. The inputs of each vehicle are its acceleration $w_i(t)$ and steering angle $\delta_{i,f}(t)$.

In this scenario, vehicles aim to select an appropriate lane to drive in to keep their desired speeds while maintaining safety. In this case, the action space of each vehicle is $\mathcal{U}_i = \{\text{Lane 1, Lane 2}\}$.

After a lane is selected, a vehicle needs to determine whether it needs to perform a lane-change, i.e., whether the selected lane (called target lane) differs from its current lane. If lane-changing is needed, then a constant steering angle $\delta_{i,f}(t) = \pm 0.9°$ (selected according to experimental data in [48]) is applied until the vehicle is in the center of the target lane, after which a lane centering controller is activated to reduce the relative heading angle between the vehicle and the road to zero. The longitudinal acceleration $w_i(t)$ of each vehicle remains zero during the process. Initially, all vehicles are placed in Lane 1, with their relative positions shown in Figure 3.

Since the lane-changing maneuver often takes around $4s$ to complete [49], the prediction horizon length T is selected to correspond to $4s$. The strategy space $\mathcal{A}_i$, $i = 1, 2, 3$, is constructed based on the action space $\mathcal{U}_i$ such that a constant action, e.g., Lane 1, is planned over the prediction horizon, i.e., $a_i(\tau) = a_i(t) \in \mathcal{U}_i$, $\forall \tau \in [t, t+T-1]$ in (17). Note that although a constant action is planned, a vehicle may change its mind after Δt, i.e., a vehicle may terminate an attempt of lane-changing and change back to its previous lane if needed. This is triggered by the receding-horizon game setting in (17).

B. Cost Function Setup

The cumulative cost of each vehicle is designed as

$$\begin{aligned} &J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}(t)) \\ &= J_i^{self}(\mathbf{a}_i(t)) + \sum_{j\in\mathcal{N}, j\neq i} J_{ij}(\mathbf{a}_i(t), \mathbf{a}_j(t)) \\ &\quad + \sum_{j\in\mathcal{N}^*_{-i}} \mathbf{1}_{\text{rules}}(\mathbf{a}_i(t), \mathbf{a}_j(t)) \end{aligned} \tag{45}$$

where the first term $J_i^{self}(\mathbf{a}_i(t))$ is to discourage lane changes, and as a result, to make lane-changing less frequent. Specifically, $J_i^{self}(\mathbf{a}_i(t))$ is designed as

$$J_i^{self}(\mathbf{a}_i(t)) = \begin{cases} 1, & \text{if } \mathbf{a}_i(t) \neq \mathbf{a}_i^*(t-1), \\ 0, & \text{otherwise.} \end{cases} \tag{46}$$

Note that $\mathbf{a}_i^*(t-1)$ represents vehicle i's selected lane at $t-1$, and $\mathbf{a}_i(t) \neq \mathbf{a}_i^*(t-1)$ indicates that a lane-changing is required to perform $\mathbf{a}_i(t)$.

The second term $\sum_{j\in\mathcal{N}, j\neq i} J_{ij}(\mathbf{a}_i(t), \mathbf{a}_j(t))$ is to avoid collision:

$$\begin{aligned} &\sum_{j\in\mathcal{N}, j\neq i} J_{ij}(\mathbf{a}_i(t), \mathbf{a}_j(t)) \\ &= \sum_{\tau=t}^{t+T-1} C_{ij}(x_i(\tau), y_i(\tau), x_j(\tau), y_j(\tau)), \end{aligned} \tag{47}$$

where

$$\begin{aligned} &C_{ij}(x_i(\tau), y_i(\tau), x_j(\tau), y_j(\tau)) \\ &= \left(\tanh\left(\beta(d_{x,c}^2 - (x_i(\tau) - x_j(\tau))^2)\right) + 1\right) \\ &\quad \cdot \left(\tanh\left(\beta(d_{y,c}^2 - (y_i(\tau) - y_j(\tau))^2)\right) + 1\right). \end{aligned} \tag{48}$$

Here $d_{x,c}$ and $d_{y,c}$ are the longitudinal and lateral collision distances, respectively, and we select $d_{x,c} = 7\ m$ and $d_{y,c} = 4.5\ m$ in the simulation. The parameter β is selected to be sufficiently large such that the $tanh$ function always takes values close to -1 or 1. The cost (48) means that at time τ, if vehicle j is a collision threat to vehicle i both laterally and longitudinally, then $J_{ij}(x_i(\tau), y_i(\tau), x_j(\tau), y_j(\tau)) = 4$; Otherwise, $J_{ij}(x_i(\tau), y_i(\tau), x_j(\tau), y_j(\tau)) = 0$. Note that the lane width of our simulation environment is set to be $5\ m$, and the lateral collision distance $d_{y,c}$ is selected to be a bit smaller than the road width such that if a vehicle is not in the same lane or in the target lane of the ego vehicle, then it is not considered as a threat to the ego vehicle.

The third term $\sum_{j\in\mathcal{N}^*_{-i}} \mathbf{1}_{\text{rules}}(\mathbf{a}_i(t), \mathbf{a}_j(t))$ is to encourage the compliance with road priority rules, which, in our simulation, are that the rear vehicle (resp. the lane-changing

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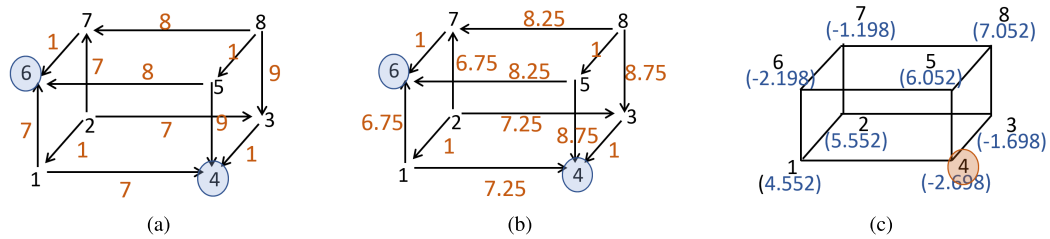

Fig. 4. (a) The flow representation of the game $\mathcal{G}$; (b) The flow representation of the potentialized game $\hat{\mathcal{G}}$; and (c) The potential function value of $\hat{\mathcal{G}}$ at each strategy profile.

vehicle) has the responsibility to take actions to avoid rear-end accidents (resp. accidents during the lane-changing). Here $\mathcal{N}^*_{-i}$ is the set of agents with higher road priority than agent i. It includes the vehicle right in front of i and the ones in vehicle i's target lane if $\mathbf{a}_i(t)$ requires lane-changing. The notation $\mathbf{1}_{\text{condition}}$ is defined as

$$\mathbf{1}_{\text{condition}} = \begin{cases} 1, & \text{if condition is true,} \\ 0, & \text{otherwise.} \end{cases} \tag{49}$$

Note that with the last term in the cost function (i.e., $\sum_{j\in\mathcal{N}^*_{-i}} \mathbf{1}_{\text{rules}}(\mathbf{a}_i(t), \mathbf{a}_j(t))$), the agents' interactions are not symmetric anymore, i.e., J_i does not satisfy the condition required in Theorem 1. We shall show in the next subsection that this formulated game is not a potential game.

C. Game Solution, Projection, and Robustness

Let us consider the formulated game:

$$\mathbf{a}^*_i(t) \in \underset{\mathbf{a}_i(t)\in\mathcal{A}_i}{\text{argmin}}\ J_i(\mathbf{a}_i(t), \mathbf{a}_{-i}(t)), \tag{50}$$

where $i \in \{1, 2, 3\}$ and J_i is given by (45). Consider the scenario where vehicles 2 and 3 drive at the same speed, faster than vehicle 1. Initially (i.e., $t = 0\ s$), all vehicles are in Lane 1 as shown in Figure 3 with safe inter-vehicle distances, i.e., $J_{ij}(\mathbf{a}_i(0), \mathbf{a}_j(0)) = 0, \forall i, j \in \{1, 2, 3\}, i \neq j$, and $\forall \mathbf{a}_i(0), \mathbf{a}_j(0) \in \mathcal{A}$. Because vehicle 3 drives faster than vehicle 1, $\Delta x_{1,3}(t) = |x_1(t) - x_3(t)|$ decreases with time. Consider the smallest $t_c > 0$ such that, if no lane-changing, then $J_{13}(\mathbf{a}_1(t_c), \mathbf{a}_3(t_c)) \neq 0$. This t_c corresponds to the earliest time such that if no action is taken, then a collision would happen at the end of the prediction horizon, $t_c + T - 1$, i.e., $\Delta x_{1,3}(t_c + T - 1) < d_{x,c}$. In our simulation, t_c approximately corresponds to $4s$, as shown in Figures 5 and 6. Let us analyze the game (50) at this key time instant. We denote this game as $\mathcal{G}$.

Since every vehicle has two possible strategies: Lane 1 or Lane 2, the total number of strategy profiles is $2^3 = 8$. For ease of representation, we label these 8 strategy profiles using the numbers between 1 and 8 as shown in Table I. In a strategy profile, the kth element represents the kth vehicle's strategy. For example, the strategy profile $(2, 2, 1)$ (which is labeled as "7" in Table I) represents that vehicles 1 and 2 select Lane 2, and vehicle 3 selects Lane 1.

With these labels, we represent the game $\mathcal{G}$ using its flow representation as shown in Figure 4(a). Here the nodes between 1 and 8 represent the 8 strategy profiles. Each edge connects two comparable strategy profiles, and the edge flow is defined by the pairwise comparison function (6). For example, the flow from the strategy profile "1" to "4" means that vehicle 3's cost will be reduced by 7 if changing from the strategy profile "1" to "4" (note that strategy profiles "1" and "4" differ only in vehicle 3's strategy). With such a flow representation, the PSNE of the game is clear: The node that has no flow "leaving" this node is a PSNE. It is because according to Definition 4, a PSNE is a strategy profile such that all its comparable strategy profiles are equally or less preferred, i.e., no cost reduction is possible by unilateral strategy deviation. With this criterion, we can observe that the game shown in Figure 4(a) has two PSNEs: The strategy profiles 4 and 6. They are marked with the blue circles.

TABLE I
STRATEGY PROFILE LABELS

Strategy profile	Label	Strategy profile	Label
$(1, 1, 1)$	**1**	$(2, 1, 2)$	**5**
$(1, 2, 1)$	**2**	$(2, 1, 1)$	**6**
$(1, 2, 2)$	**3**	$(2, 2, 1)$	**7**
$(1, 1, 2)$	**4**	$(2, 2, 2)$	**8**

The strategy profile 4 corresponds to (1,1,2), representing that vehicle 3 moves to Lane 2 while the other two vehicles stay in Lane 1. This NE is found in our simulation when we use the best response algorithm (i.e., Algorithm 1) to solve the game. The vehicles' behavior at three important time instants is illustrated in Figure 5. The strategy profile 6 corresponds to (2,1,1), representing that vehicle 1 moves to Lane 2 while the other two vehicles stay in Lane 1. This equilibrium is also found in our simulation by the best response algorithm, and its corresponding vehicles' behavior can be visualized in Figure 6. Since both strategy profiles 4 and 6 are PSNEs, they are both individually optimal solutions and are both able to resolve the potential collision conflict between vehicles 1 and 3 with the minimum lane-changing effort. However, we also note that these two PSNEs are not equally preferred: Considering the road priority rule, vehicle 3 should perform the lane-changing to avoid the rear-end accident, resulting in that strategy profile 4 is preferable to 6 considering the global situation.

Next we project the game $\mathcal{G}$ in Figure 4(a) onto the potential game space using Algorithm 3. The potentialized game (denoted as $\hat{\mathcal{G}}$) has the flow representation as shown in Figure 4(b). It is clear that the flows in Figure 4(b) do not coincide with that in Figure 4(a), indicating that the original game is not a potential game.

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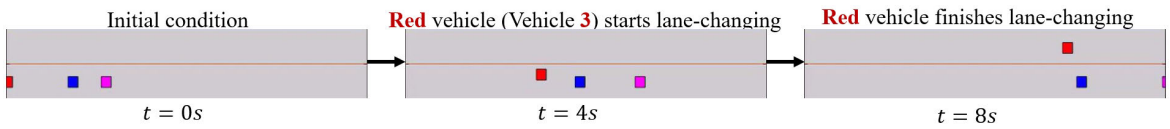

Fig. 5. Vehicle lane-changing behavior with the strategy profile 4, a preferable PSNE.

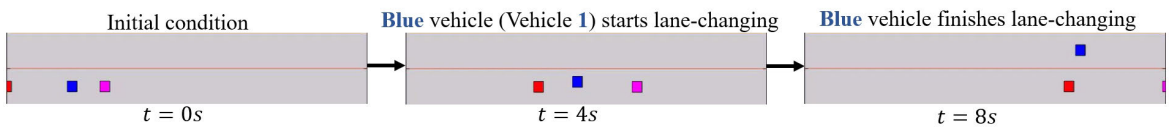

Fig. 6. Vehicle lane-changing behavior with the strategy profile 6, a PSNE.

For the projected game $\hat{\mathcal{G}}$, it also has two PSNEs: Strategy profiles 4 and 6. For each PSNE, according to Theorem 2, its robustness margin can be calculated as

$$\mu(\hat{\mathbf{a}}^*) = \frac{1}{2} \min_{i \in \mathcal{N}} \min_{(\mathbf{a},\mathbf{b}) \in \mathcal{C}_i, \mathbf{a} \neq \mathbf{b}} \left(\hat{J}_i(\mathbf{b}) - \hat{J}_i(\mathbf{a}) \right) = 0.5 \tag{51}$$

By calculating $\Delta J_i = J_i - \hat{J}_i$, where $\hat{J}_i$ is the cost of the potentialized game derived by Algorithm 3, we can find that the game projection leads to $\|\Delta J\|_\infty = 0.125$. Because $\|\Delta J\|_\infty < \mu(\hat{\mathbf{a}}^*)$, we know that such a game projection would not lead to a solution deviation, i.e., all the PSNE of $\hat{\mathcal{G}}$ remain a PSNE of $\mathcal{G}$. This result is confirmed by observing the PSNE of the games from their flow representations in Figures 4(a) and 4(b), respectively. Therefore, although the original game is not a potential game and that the potentialized game differs from the original game (i.e., cost deviations exist), the sets of solutions of the two games coincide.

The potential function F of the potentialized game $\hat{\mathcal{G}}$ associated with each strategy profile is shown in Figure 4(c). We can observe from this figure that the PSNE that minimizes the potential function is unique and is the strategy profile 4 (marked with the orange circle). Therefore, strategy profile 4 is not only individually optimal (in the sense of NE) but also socially optimal (in the sense of optimizing the "social interest" characterized by the potential function), consistent with our analysis above that strategy profile 4 is preferable to 6. Moreover, to solve the potentialized game $\hat{\mathcal{G}}$, one can use the potential function optimization algorithm (i.e., Algorithm 2), which requires one optimization as opposed to iterative optimizations in the best response algorithm (i.e., Algorithm 1). The above analysis confirms the merits of the potential/potentialized game.

We also report the running time of the game projection. The running time is collected from MATLAB® on a laptop with an Intel Core i9-12900H processor clocked at 2.50 GHz and with 32 GB of RAM. For each game projection, i.e., each execution of Algorithm 3 to find the potentialized game $\hat{\mathcal{G}}$ and the associated potential function F, the average running time is $0.015s$, confirming that it's practical to use game projection for real-time operation.

D. More Complex Scenarios

We next increase the complexity of the case study and consider a 5-vehicle scenario as shown in Figure 7. We let the five vehicles' speeds and initial positions be randomly selected to test the algorithm effectiveness in various scenarios. We observe that in all generated situations, the potential game based framework enables safe and efficient lane-changing maneuvers such that all vehicles can keep their desired speeds while maintaining safety, confirming the effectiveness of the proposed game-theoretic approach. The simulation animation videos are available at *https://youtu.be/BHmX-ldwWOU*.

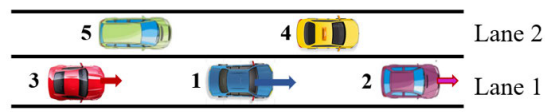

Fig. 7. Highway five-vehicle lane-changing scenario.

With the number of vehicles increasing from 3 to 5, the average running time of each game projection increases from $0.015s$ to $0.022s$. To better understand the scalability, we also tested the cases when 10 vehicles are involved (i.e., a 10-player game). In all tested scenarios, the vehicles are able to keep safe while tracking their respective desired speeds, confirming the algorithm effectiveness in complex scenarios. The average running time of each game projection is $0.06s$, confirming the scalability for on-line use.

VII. Conclusion

To make game-theoretic approaches practical for autonomous driving, challenges caused by game complexity and incomplete information have to be addressed. Potential games, thanks to their appealing properties, can address the game complexity challenge. However, conditions to construct a potential game can be restrictive. To relax the conditions, a game projection technique was employed, from which, one may project an arbitrary game onto the potential game space and then apply the potential game technique. Such a projection may lead to cost function deviations, but may not necessarily lead to a solution deviation. To answer whether and when a solution deviation happens, as well as to address the incomplete information challenge, we studied the robustness margin of a game solution, which was defined as the maximum allowed cost function deviations that would not change the optimality of the game solution. Closed-form robustness margin expressions were derived to inform the trustworthiness of a game solution and to facilitate game designs. Simulation case studies showed that asymmetric cost

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functions that reflect priority rules of the road can be handled while the projection can be performed within a reasonable computational time. Future studies will focus on providing theoretic safety guarantees with recursive feasibility, more computationally efficient algorithm design, and verification and validation on real-world roads.

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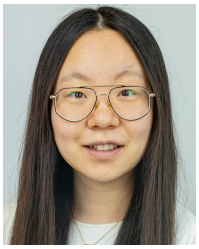

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